

Rayan Mohammed

Kochi, Kerala, India | (+91) 7012858326 | rayanhafees@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFILE

Computer Science graduate with experience building backend systems, REST APIs, workflow automation, and scalable software solutions. Strong foundation in distributed systems, databases, cloud infrastructure, and software engineering principles, with hands-on experience developing production-oriented applications and automation workflows.

SKILLS

Programming	Python, Java, JavaScript, SQL
Backend	Django, REST API Development, MySQL
Tools	Git, GitHub, Linux, Docker, Docker Compose, n8n
Cloud	Amazon Web Services (AWS) (EC2, S3, IAM, VPC, Auto Scaling, ELB)
Data & Analytics	Pandas, NumPy, Scikit-learn
Core Concepts	Data Structures, OOP, DBMS, Distributed Systems

EXPERIENCE

CVS Info Solutions, Infopark
Software Development Intern

July 2024 – September 2024

- Debugged REST API endpoints and backend modules, resolving issues in authentication, data flow, and user access
- Implemented and analyzed role-based access control (RBAC) for secure user and admin workflows
- Investigated API failures using logs and improved database query performance
- Collaborated with developers to reproduce, diagnose, and fix functional and UI defects
- Documented issues and solutions to streamline debugging and testing workflows

MicroIT – Student Tech Initiative
Co-Founder

December 2023 – January 2025

- Led backend development for web and mobile solutions, focusing on API design and system architecture
- Built and deployed core features for a B2B travel platform (VayoAura)
- Translated client requirements into scalable workflows and backend modules
- Handled debugging, testing, and iteration cycles for production-ready applications

AIESEC in Amaravati
Business Development Intern

August 2022 – May 2023

- Ran outbound prospecting across email, LinkedIn, and cold calls, engaging 40–60 B2B prospects per week
- Pitched tailored value propositions to decision-makers, handled objections, and qualified leads based on fit and intent
- Scheduled discovery calls and partner meetings for senior team members, booking 5–6 meetings per month on average
- Maintained a structured lead pipeline with disciplined follow-ups, ensuring consistent engagement across multi-touch sequences

PROJECTS

Task Scheduling Optimization in Distributed Systems Python, PyTorch, NumPy, Alibaba Cluster Trace Dataset

- Designed a hybrid DQN + NSGA-II + Honey Badger Optimization algorithm for multi-objective task scheduling on real Alibaba cluster traces (100 processors, 100–500 task DAGs)
- Built a Deep Q-Network manager that adaptively selects among 7 evolutionary operators (5 HBO moves, crossover, memetic local search) using experience replay and ϵ -greedy exploration
- Achieved HEFT-optimal makespan on a 5-job concurrent workload (450 tasks, 10 processors); outperformed GA, PSO, GWO, EHO, HBO, and RL baselines by 38–44% with $p < 0.001$ (Wilcoxon rank-sum, 30 independent runs)
- Validated multi-objective convergence across ZDT (1–6) and DTLZ (1–7) benchmark suites, producing well-spread Pareto fronts with up to 50 non-dominated solutions per configuration
- Engineered an end-to-end DAG-based scheduling simulator with HEFT priority ranking, SLA deadline enforcement, and joint optimization of makespan, energy, wait time, and processor utilization

Clara AI Automation Pipeline

Docker, Docker Compose, n8n, JSON, Workflow Automation

- Built a containerized automation pipeline using Docker Compose and n8n to process AI call transcripts and generate structured agent configuration artifacts
- Designed version-controlled workflow pipelines for onboarding updates, automated data extraction, and JSON-based account memo generation
- Implemented modular orchestration workflows for transcript ingestion, transformation, and artifact generation with local containerized deployment
- Improved reproducibility and debugging through isolated service deployment and structured workflow automation

VSync – Smart College Companion Platform

Flutter, Dart, REST API

- Developed a modular full-stack academic management platform using Flutter and REST APIs
- Designed structured data models for academic records and GPA computation
- Integrated backend services for dynamic data retrieval and state management
- Implemented scalable UI components with clean architecture principles

Gesture-Controlled Drone

Raspberry Pi, OpenCV, Python

- Developed a real-time computer vision system to map hand gestures to drone flight controls
- Implemented gesture recognition pipelines using OpenCV for accurate motion detection
- Optimized image processing for low-latency response
- Designed for surveillance, search operations, and hands-free filming use cases

EDUCATION

Vellore Institute of Technology Andhra Pradesh (VIT-AP)

2022 – 2026

B.Tech in Computer Science Engineering

CERTIFICATIONS

- [AWS Academy Cloud Architecting \(2024\)](#)
- [AWS Academy Cloud Foundations \(2024\)](#)
- [CouchDB Distributed Database System \(2024\)](#)